

European Championship in Trustworthy AI

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Title: Member vs Generated Inference (MGI)

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Task Overview

Image generative models (IGMs) are trained on data scraped from the internet, much of which was itself produced by earlier generative models. This creates data circuits where an IGM is fine-tuned on generated samples, so the boundary between natural training, test, and generated data becomes blurred.

Consider a detector that assigns each sample to one of three categories: **M** (member training samples), **N** (non-member samples from the same distribution as the training set), or **G** (model-generated samples). Your goal is to manipulate the detector's predictions so that it makes the following targeted errors: $\mathbf{M} \rightarrow \mathbf{N}$, $\mathbf{M} \rightarrow \mathbf{G}$, $\mathbf{N} \rightarrow \mathbf{M}$, $\mathbf{N} \rightarrow \mathbf{G}$, $\mathbf{G} \rightarrow \mathbf{M}$, $\mathbf{G} \rightarrow \mathbf{N}$, where $A \rightarrow B$ denotes a sample from class A being classified as class B.

What you have access to:

- Image generative model (RAR).
- A dataset of 900 images evenly distributed across three classes (M, N, and G), with 300 images per class.

What you do not have access to:

- The detector to classify samples as being **M**, **N**, or **G**.

Evaluation Metric

The detector is based on 2 objectives:

- Membership Inference Attack (MIA) score: detect if an image is a training sample of the given model.
- Generated image attribution: detect if an image is generated by the given model.

Your performance is based on:

- Misclassification by the detector.
- Image Quality Measure (Mean Squared Error).

You will submit images for evaluation. If a submitted image induces the correct targeted misclassification direction, it passes the detector check and receives $\text{DetectorScore}=1$; otherwise, it receives the $\text{DetectorScore}=0$. For each passing image, its quality is measured as $(1 - \text{MSE})$, where MSE is the normalized mean squared error between the submitted image and the original reference image (the lower the MSE the better). The final score is the mean over all $n=1800$ samples that you sent to us:

$$score = \frac{1}{n} \sum_{i=1}^n DetectScore \times (1 - MSE_i)$$

The score is bounded in [0, 1]. A higher score means more images fool the detector while preserving visual quality. The maximum score of 1.0 is achieved when every image passes with zero pixel change; 0.0 means no image is misclassified.

Dataset

A single archive (Dataset.zip) is provided containing 900 images with the following naming convention:

- img_000.png - img_299.png: subset M
- img_300.png - img_599.png: subset N
- img_600.png - img_899.png: subset G

The RAR generative model can be downloaded from [this](#) hugging face repository.

[hackathon_setup.sh](#) provided to you creates your team folder, downloads the datasets and creates a per task environment for you.

OR

Dataset can be downloaded as follows:

`cd your-folder`

`wget https://huggingface.co/datasets/SprintML/MGI`

Additional Resources

You are supposed to submit your scores using the code file [submission_template.py](#) provided. Remember to replace `YOUR_API_KEY_HERE` with your actual API-key (keep the double quotes), and replace `/PATH/FILE.npz` with your real results.

Submission Format

The submission file should be a binary .npz compressed file of your modified images that contains image IDs from 0000 to 1799.

Requirements

- Each image must appear exactly once
- All of the images should be included in the compressed file
- All of the images must be in the shape of 256 x 256 x 3
- You should submit the images with the same IDs as the naming convention:
 - 0000 to 0299 for **M** → **N**,
 - 0300 to 0599 for **M** → **G**,
 - 0600 to 0899 for **N** → **M**,
 - 0900 to 1199 for **N** → **G**,
 - 1200 to 1499 for **G** → **M**,
 - 1500 to 1799 for **G** → **N**

Submissions that do not meet these requirements will be rejected automatically.

Scoring

All submissions are automatically validated and evaluated on the hidden test set.

- Public leaderboard: based on 30% of data
- Private leaderboard: final ranking based on the remaining 70%

A higher score indicates more data points that fooled the detector and preserve the quality of the image.

Leaderboard

After evaluation, your results can be found in the leaderboard:

- You can access the leaderboard for this task at http://35.192.205.84/leaderboard_page. This will help you to compare your solutions with other teams and see where you stand.
- The leaderboard shows the best result per team only. As output to your request, you will get back the score for your current submission. If it is lower than the score saved in the leaderboard, the score will not be updated.

References:

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2. “Data Provenance for Image Auto-Regressive Generation” Bihe Zhao, Louis Kerner, Michel Meintz, Tameem Bakr, Franziska Boenisch, Adam Dziedzic ICLR 2026
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3. “AEDR: Training-Free AI-Generated Image Attribution via Autoencoder Double-Reconstruction” Chao Wang, Zijin Yang, Yaofei Wang, Weiming Zhang, Kejiang Chen AAAI 2026 <https://arxiv.org/abs/2507.18988>
4. “Membership Inference Attacks against Machine Learning Models” Reza Shokri, Marco Stronati, Congzheng Song, Vitaly Shmatikov S&P 2017
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6. “Privacy Attacks on Image AutoRegressive Models” Antoni Kowalczyk, Jan Dubiński, Franziska Boenisch, Adam Dziedzic, ICML 2025
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